

Practical Exercises With FRESCO

The object of these exercises is to give you practical experience in running FRESCO for a variety of cases, the aim being to build confidence by an easy progression from simple optical model inputs, through coupled channels for inelastic scattering to DWBA and CRC calculations for transfer reactions. When you have worked through the complete set you should be able to assemble an input file for – almost – any reaction covered by these models.

1. Optical Model Calculations

We begin with the simplest case, one-shot optical model calculations using a given set of potential parameters. Find the input file p12C.fri in the materials provided. Create a new directory and copy this file to it. Run FRESCO with it by typing the following command:

prompt:fresco<p12C.fri>p12C.fro &

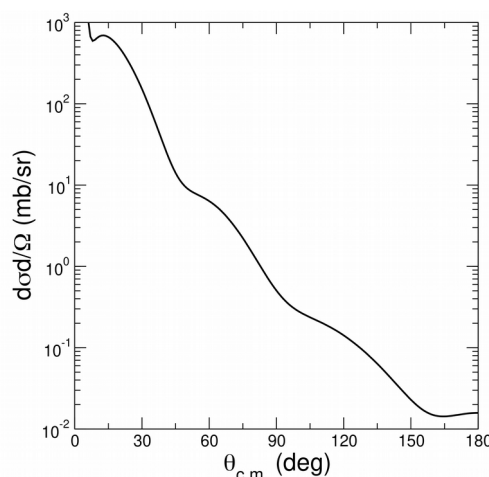
This will create a number of files, p12C.fro and several called fort.XX. Of the latter, only fort.56 and fort.201 are of interest to us at the present time. This input will take less than 1 second to run, so it is not really necessary to run in background mode. However, it is good practice to do so since more complicated inputs can take several hours to complete, or even days in extreme cases.

Open the main output file, p12C.fro, and scroll down it. You should see repeated the input optical potential parameters and other information, as per the example in the lectures. You will eventually come to the calculated total reaction cross section, which should appear as follows:

CUMULATIVE REACTION cross section = 269.14864

This value is the reaction cross section in mb, a useful quantity in its own right and as an additional check when repeating optical model calculations from the literature (in the latter case it is unlikely that you will obtain *exactly* the same as the published value, particularly if it is an older article, due to differences in machine precision, choice of mass values etc. However, if you have done things correctly you should obtain a number that is very close to the published value, misprints excepted!).

Next, open the file fort.201. You will see some initial commands, followed by three columns of figures; these are the centre of mass scattering angle in degrees, the elastic scattering cross section in mb/sr and the first rank (or vector) analysing power, iT_{11} . The initial commands enable one to plot the file directly using xmgrace. If you are not familiar with this program you may easily edit the file to give the listing of the angular distribution(s) and plot these with your usual plotting program. The cross section angular distribution should look like this:



If you open the file fort.56 you should see the following:

```
# Fusion cross sections for ELAB= 49.500, ECM= 45.665
0.5 17.88 17.88 0.000
1.5 33.63 33.63 0.000
2.5 49.07 49.07 0.000
3.5 57.42 57.42 0.000
4.5 47.84 47.84 0.000
5.5 29.86 29.86 0.000
.
.
.
20.5 0.3178E-03 0.3178E-03 0.000
&
```

This file gives the breakdown of the reaction cross section (and absorption and outgoing cross sections for more complicated calculations) as a function of partial wave. The first column is the partial wave (J) value in units of $\hbar$, the second column the absorption cross section in mb, the third column the reaction cross section and the fourth column the outgoing cross section (here identically zero since we do not have any explicit couplings, all absorption being due to the imaginary part of the optical potential, represented by column 2). This information is also given in p12C.fro but fort.56 gives it in a form convenient for plotting. It provides a useful check that sufficient partial waves have been allowed for in the input; as a rough guide, for optical model calculations if the reaction cross section for the final partial wave is around 0.1E-01 or less then you will certainly have enough partial waves. We will return to this subject when we deal with coupled channels (CC) calculations.

We will now proceed to modify the input file to include a surface imaginary term. Create a new working directory and copy the file p12C.fri to it. Rename it p12CKD.fri (the KD denotes Koning and Delaroche, since we will modify the input to calculate the elastic scattering using the global proton parameters of A. J. Koning and J. P. Delaroche, Nucl. Phys. A **713**, 231 (2003). The definition of the optical potential in this work is as follows:

$$\begin{aligned}\mathcal{U}(r, E) = & -\mathcal{V}_V(r, E) - i\mathcal{W}_V(r, E) - i\mathcal{W}_D(r, E) \\ & + \mathcal{V}_{SO}(r, E).\mathbf{l}.\sigma + i\mathcal{W}_{SO}(r, E).\mathbf{l}.\sigma + \mathcal{V}_C(r),\end{aligned}$$

where:

$$\begin{aligned}\mathcal{V}_V(r, E) &= V_V(E) f(r, R_V, a_V), \\ \mathcal{W}_V(r, E) &= W_V(E) f(r, R_V, a_V), \\ \mathcal{W}_D(r, E) &= -4a_D W_D(E) \frac{d}{dr} f(r, R_D, a_D), \\ \mathcal{V}_{SO}(r, E) &= V_{SO}(E) \left(\frac{\hbar}{m_\pi c} \right)^2 \frac{1}{r} \frac{d}{dr} f(r, R_{SO}, a_{SO}), \\ \mathcal{W}_{SO}(r, E) &= W_{SO}(E) \left(\frac{\hbar}{m_\pi c} \right)^2 \frac{1}{r} \frac{d}{dr} f(r, R_{SO}, a_{SO}).\end{aligned}$$

and:

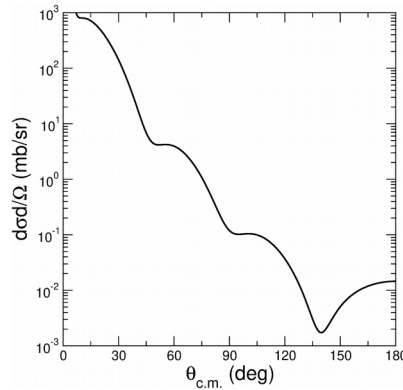
$$f(r, R_i, a_i) = \left(1 + \exp[(r - R_i)/a_i]\right)^{-1}$$

with $R_i = r_i A^{1/3}$ and the usual form of Coulomb potential. Compare this form carefully with the definitions in the FRESKO manual (section 2.4, card types 9 and 10) to ensure that you enter the parameters correctly. N.B. $\sigma = 2s$, where s is the intrinsic spin of the projectile (1/2 here for the proton). Since the parameters of the global potential are calculated using quite complicated expressions, to save time the values for protons (the potential also covers neutron scattering) scattering from a ^{12}C target at an incident energy of 49.5 MeV are given below:

$V_V = 38.2$ MeV, $W_V = 5.55$ MeV, $r_V = 1.13$ fm, $a_V = 0.68$ fm,
 $W_D = 4.37$ MeV, $r_D = 1.31$ fm, $a_D = 0.52$ fm,
 $V_{SO} = 4.73$ MeV, $W_{SO} = -0.36$ MeV, $r_{SO} = 0.90$ fm, $a_{SO} = 0.59$ fm,
 $r_C = 1.54$ fm.

N.B. this potential contains both volume and surface imaginary terms, so think carefully how this is implemented. For this calculation you should not need to change any of the integration parameters (card 1) or the number of partial waves.

If all goes well you should obtain a cumulative reaction cross section of 390.98630 mb and an angular distribution that looks like this:



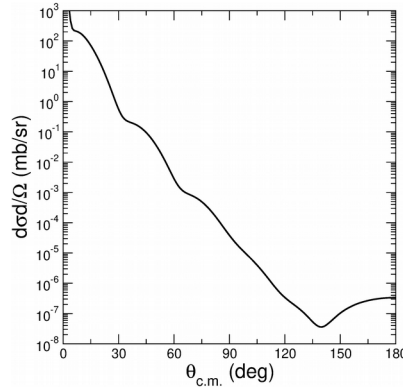
You should now be familiar with how to include surface terms in the optical potential.

For our next exercise, we stay with protons incident on ^{12}C and the Koning and Delaroche (K-D) global potential but we now increase the incident energy to 200 MeV. Create another new working directory and copy the file p12CKD.fri to it. Rename the file p12CKD200.fri. The potential parameters will be somewhat different since the K-D global potential is energy dependent. The new values are as follows:

$V_V = 12.4$ MeV, $W_V = 13.1$ MeV, $r_V = 1.13$ fm, $a_V = 0.68$ fm,
 $W_D = 0.17$ MeV, $r_D = 1.31$ fm, $a_D = 0.52$ fm,
 $V_{SO} = 2.59$ MeV, $W_{SO} = -1.95$ MeV, $r_{SO} = 0.90$ fm, $a_{SO} = 0.59$ fm,
 $r_C = 1.54$ fm.

Run the new input file. If you do not get the result you expect, check the main output file and think about what it tells you in the light of what we learned yesterday in the lecture on optical model calculations. Remember that we have changed significantly the incident energy, so we may need to modify some of the “control” parameters of our input such as the number of partial waves (JTMAX), the step size (HCM) etc. The main output file will give you some clues as to what, if anything, needs to be changed and, in some cases, the values required.

You should obtain a cumulative reaction cross section of 191.39878 mb and an angular distribution that looks like this:

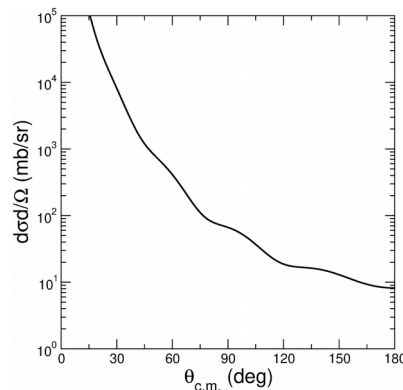


You should begin to have some idea of what to be aware of in the – apparently – almost trivial operation of using FRESKO to perform an optical model calculation. We shall look at two further examples with protons as projectile, this time scattering from a ^{208}Pb target.

We will consider two different incident proton energies, 16 and 200 MeV. The object of this exercise is to give you experience in how the requirements of a successful optical model calculation vary with mass as well as energy. We begin as always by creating a new working directory – good practice since otherwise FRESKO will over-write the fort.XX files, erasing information we want to keep – and copying the file p12CKD.fri to it; we shall continue to use the K-D parameters. Rename it p208PbKD.fri. The new potential parameters for 16 MeV protons on ^{208}Pb are as follows:

$$\begin{aligned} V_V &= 49.3 \text{ MeV}, W_V = 0.90 \text{ MeV}, r_V = 1.24 \text{ fm}, a_V = 0.65 \text{ fm}, \\ W_D &= 10.24 \text{ MeV}, r_D = 1.25 \text{ fm}, a_D = 0.63 \text{ fm}, \\ V_{SO} &= 6.01 \text{ MeV}, W_{SO} = -0.06 \text{ MeV}, r_{SO} = 1.08 \text{ fm}, a_{SO} = 0.59 \text{ fm}, \\ r_C &= 1.22 \text{ fm}. \end{aligned}$$

Take a value of 207.9767 for the target mass on card 6; don't forget to change A#1 on card 9 too! You should obtain a cumulative reaction cross section of 937.73449 mb and an angular distribution that looks like this:

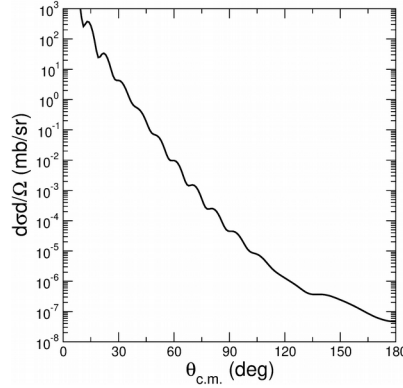


Even if your input file appears to run with no problems be sure to check that your integration parameters are reasonable (JTMAX, RMATCH etc.).

Now repeat the exercise for 200 MeV protons on ^{208}Pb . As always, create a new directory, copy p208PbKD.fri to it and rename it as p208PbKD200.fri. The potential parameters have changed slightly:

$$\begin{aligned} V_V &= 8.00 \text{ MeV}, W_V = 13.98 \text{ MeV}, r_V = 1.24 \text{ fm}, a_V = 0.65 \text{ fm}, \\ W_D &= 0.48 \text{ MeV}, r_D = 1.25 \text{ fm}, a_D = 0.63 \text{ fm}, \\ V_{SO} &= 2.88 \text{ MeV}, W_{SO} = -1.93 \text{ MeV}, r_{SO} = 1.08 \text{ fm}, a_{SO} = 0.59 \text{ fm}, \\ r_C &= 1.22 \text{ fm}. \end{aligned}$$

As before, if you do not get the result you are expecting, look at the main output file (and fort.56) and think about what you see there. Remember that we have increased the incident proton energy considerably, so you may need to modify some of the “control” parameters again. If all is well, you should have a cumulative reaction cross section of 1624.10290 mb and an angular distribution that looks like this:



We have now finished with proton projectiles and will proceed to look at deuterons. Since the deuteron intrinsic spin is 1.0 (compared to 0.5 for the proton) this gives us the opportunity to address the slightly non-standard formulation of the spin-orbit term in FRESCO. For our example we shall consider 52 MeV deuterons scattering from a ^{58}Ni target. We shall again use a set of global potential parameters, this time that of W. W. Daehnick et al., Phys. Rev. C **21**, 2253 (1980), specifically adapted to deuteron scattering. The definition of the potential used in this paper is as follows:

$$\begin{aligned} V(r) &= -V_R f(r, r_0, a_0) + i4a_I W_D \frac{d}{dr} f(r, r_I, a_I) \\ &\quad - iW_S f(r, r_I, a_I) \\ &\quad + V_{LS} \left(\frac{\hbar}{m_p c} \right)^2 (\vec{L} \cdot \vec{S}) \frac{1}{r} \frac{d}{dr} f(r, r_{LS}, a_{LS}) + V_{Coul}, \end{aligned}$$

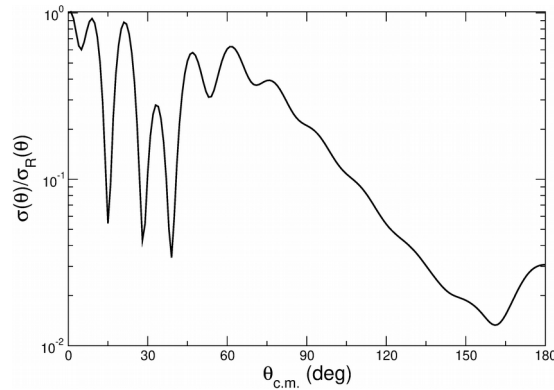
where the Woods-Saxon well f is given by

$$f(r, r_i, a_i) = \left[1 + \exp\left(\frac{r - r_i A^{1/3}}{a_i}\right) \right]^{-1};$$

Note that the spin-orbit term is defined in terms of $\mathbf{L} \cdot \mathbf{S}$ where $\mathbf{S}$ is the intrinsic spin of the deuteron. For 52 MeV deuterons incident on a ^{58}Ni target the parameters, *according to the above definition*, are as follows:

$$\begin{aligned} V_R &= 81.3 \text{ MeV}, \quad r_0 = 1.17 \text{ fm}, \quad a_0 = 0.797 \text{ fm}, \\ W_S &= 3.21 \text{ MeV}, \quad W_D = 10.34 \text{ MeV}, \quad r_I = 1.325 \text{ fm}, \quad a_I = 0.786 \text{ fm}, \\ V_{LS} &= 5.82 \text{ MeV}, \quad r_{LS} = 1.07 \text{ fm}, \quad a_{LS} = 0.66 \text{ fm}, \\ r_C &= 1.30 \text{ fm}. \end{aligned}$$

Begin as always with a new directory, copy p12C.fri to it and rename d58Ni.fri. Change the input to suit 52 MeV d + ^{58}Ni with the above parameters, *taking particular care to compare the form of the potential with that used in FRESKO*. You may take the mass of ^{58}Ni as 57.9353. Also take care to make any necessary adjustments to “control” parameters such as HCM, RMATCH and JTMAX. Arrange the input file so that FRESKO outputs the differential cross section angular distribution expressed as a ratio to the Rutherford cross section rather than in absolute units (check the FRESKO manual under card type 3, variable THMAX). You should obtain a cumulative reaction cross section of 1714.10009 mb and an angular distribution as follows:



Hint: if your angular distribution does not look quite like this, check carefully the definition of the spin-orbit term.

Finally, we consider a heavy-ion scattering system, $^7\text{Li} + ^{208}\text{Pb}$. The Coulomb term of the optical potential is naturally more important for heavy ions compared to protons, deuterons and other light ions (conventionally any projectile up to and including the α particle) but otherwise the potentials are usually of exactly the same basic form as for light ions (one minor exception being the definition of the radius parameter in the Woods-Saxon potential, the full radius often being defined in terms of the sum of the cube roots of the projectile and target mass numbers for heavy ions). The increased importance of the Coulomb term will impact on the requirements for some of the “control” parameters like HCM, RMATCH and JTMAX, the latter two in particular.

We will calculate the elastic scattering for 35 MeV ^7Li ions incident on a ^{208}Pb target. Create a new directory and copy the file p208PbKD.fri to it and rename it 7Li208Pb.fri (this time we will modify the input for protons on ^{208}Pb since it already has the target we need). Unusually for heavy ions, there is a rather good set of global optical model parameters for ^7Li scattering by J. Cook, Nucl. Phys. A **388**, 153 (1982) so we shall use it in our example. The form of the potential is standard Woods-Saxon for both real and imaginary parts, plus the usual Coulomb term. The parameters are:

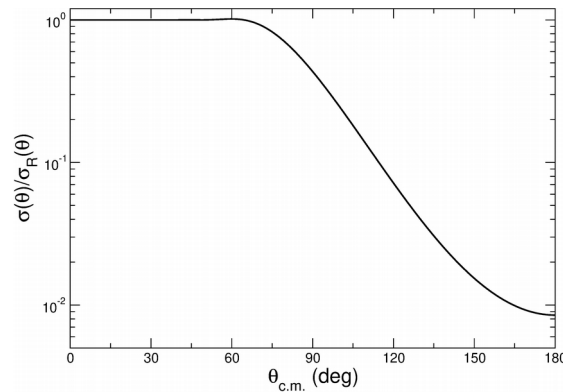
$$V_R = 114.2 \text{ MeV}, r_0 = 1.286 \text{ fm}, a_0 = 0.853 \text{ fm},$$

$$W_S = 40.13 - 0.341A + 0.00093A^2 \text{ MeV}, r_I = 1.739 \text{ fm}, a_I = 0.809 \text{ fm},$$

$$r_C = 1.30 \text{ fm}$$

where A is the mass number of the target, 208 here, and the radius convention is $R_i = r_i \times A^{1/3}$ fm. Note that there is no physical significance in the convention used to calculate the actual radius parameter of the Woods-Saxon potential, one simply needs to make sure which one has been used in order to ensure that the potential is calculated correctly. You may take the mass of ${}^7\text{Li}$ as equal to 7.0160 and the spin-parity of its ground state as $3/2^-$ (this will not affect the result in this case, since there are no spin-dependent terms in the calculation, but it is good practice to use the correct value).

The object of this exercise is to pull together what you have learned so far. You should arrange your input file to output the differential cross section angular distribution expressed as a ratio to the Rutherford cross section (standard for heavy ion scattering). Check carefully the outputs to ensure that you have adequate “control” parameter values. You should obtain a cumulative reaction cross section of 957.94403 mb and an angular distribution that looks like this:



This brings to an end the first set of exercises and you should now be able to calculate an elastic scattering angular distribution from a given set of optical potential parameters. In the next section we will look at how to perform parameter searches to fit data using the optical model.

2. Optical Model Fitting.

For our first exercise in fitting data we shall return to our 49.5 MeV $p + {}^{12}\text{C}$ example. Go to the working directory you created for this example in the previous set of exercises and create a sub-directory called “search”. Copy the file p12CKD.fri to it. Locate the files sf.inp and search.in in the materials provided and copy them to the “search” directory. Unless you are familiar with the MINUIT χ^2 minimisation routines you should not need to edit sf.inp in any of these exercises.

Open the file search.in. It should look as follows:

```
'p12CKD.fri' 'p12CKD.fro' 6 1
&variable kind=1 name='V' step=0.1 kp=1 pline=2 col=1 /
&variable kind=1 name='r' step=0.1 kp=1 pline=2 col=2 /
&variable kind=1 name='a' step=0.1 kp=1 pline=2 col=3 /
&variable kind=1 name='W' step=0.1 kp=1 pline=2 col=4 /
&variable kind=1 name='rw' step=0.1 kp=1 pline=2 col=5 /
&variable kind=1 name='aw' step=0.1 kp=1 pline=2 col=6 /
&data type=0 delta=0 iscale=2 idir=0 lab=F abserr=T/
10.88 0.76100E+03 0.10654E+02
```

```
13.05 0.75900E+03 0.11385E+02
15.22 0.70900E+03 0.10635E+02
```

```
.
```

See Appendix VI of the FRESCO manual for a full description of the various components that go to make up search.in. Here we will give a brief summary of the most important. The first line defines the input and output files that will be used with FRESCO as part of the searching process carried out by SFRESCO. The “6” denotes the number of parameters to be searched on (simultaneously). Note that it is not usually good practice to search on more than about 2-3 parameters of an optical potential simultaneously; this input is merely an example. The “1” following it denotes the number of data sets which we wish to fit, in this case just one (the cross section angular distribution).

The following 6 lines define the variables to be searched on, e.g.:

```
&variable kind=1 name='V' step=0.1 kp=1 pline=2 col=1 /
```

The ampersand (& character) merely makes this line active, so to save editing, if one wished to search on just this single parameter it is possible to do so using this input merely by removing the ampersands from the beginning of the other 5 lines, although don’t forget to change the ‘6’ to a ‘1’ too; variable kind=1 merely labels this as an optical potential variable (see Appendix VI for other types of variable it is possible to search on); name='V' is just the name given to this variable (you may use any label you wish); step=0.1 is a control parameter for the step size used in the searching procedure, the default value is 0.01 and this is fine for most purposes.

The following 3 variables, kp, pline and col, are particular to variable kind=1 and define the position of the parameter in the FRESCO input file (p12CKD.fri here): kp=1 is the index of the potential, as on card types 9 and 10, pline=2 defines the line of the potential where the variable is found, here the second line (starting from the initial card type 9 where the mass(es) and Coulomb radius parameter are defined), finally col=1 defines the position of the parameter to be searched on on this line, in this case column 1 corresponding to variable P1 on card type 10 (real potential depth in this case).

The line:

```
&data type=0 delta=0 iscale=2 idir=0 lab=F abserr=T/
```

merely defines how the data to be fitted are formatted, see Appendix VI for details. There follow the data in format: scattering angle, differential cross section in mb/sr, absolute error of differential cross section in mb/sr.

After this brief recap try searching on the potential parameters of p12CKD.fri to obtain a better fit to the data. To begin with, try searching on individual parameters such as V_V and W_V (or W_D). To run sfresco, type the following command:

prompt:sfresco<sf.inp

This will default output to the screen, the most useful way to run sfresco since it enables one to see how the fitting is progressing. In this case, the searches will only take a few seconds or less (unless you decide to search on all 15 parameters simultaneously – not recommended!). Scrolling back through the output, you should see something like this:

SFRESKO - FRES 3.2: Search Coupled Reaction Channels

Please give search file name
SFresco search file = search.in

Fresco input file = p12CKD.fri
and output file = p12CKD.fro

Search on 1 variables for 1 datasets,
MINUIT RELEASE 96.03 INITIALIZED. DIMENSIONS 100/ 50 EPSMAC= 0.89E-15

Define SEARCH VARIABLES:

Variable 1=V is potential KP= 1, line= 2 col= 1
value from Fresco input, step 0.1000, min,max 0.0000 0.0000

Read definition of data set 1
Read DATA SET 1 from file =
of type 0 in mb with absolute errors, CM for state/excit 1 1, idir= 0

Angular cross section T00

Read data set 1

54 data points:

Angle CM	Datum	Absolute error
10.880	761.0	10.65
13.050	759.0	11.38
.	.	.
.	.	.
.	.	.
143.220	0.4110E-01	0.1315E-02
146.940	0.3630E-01	0.1561E-02

FRESKO output to p12CKD.fro*

Calculate scattering at energies
49.500

Var 1=V value 38.200001

Total ChiSq/N = 688.8897 from 688.890

PARAMETER DEFINITIONS:

NO.	NAME	VALUE	STEP SIZE	LIMITS
1	'V	38.200	0.10000	no limits

sfresco>

Thus far the output is merely repeating the information you gave SFRESKO. It lists the data you are fitting, the variable(s) you are searching on, their initial values and the initial χ^2 per point value (here 688.890, so hardly a good fit ...). It now passes to MINUIT to perform the search:

Call MINUIT

with NOGradient, STRategy=0, ERRordef= 0.0185

** 1 **SET NOGRADIENT

** 2 **SET STRAT 0.000

NOW USING STRATEGY 0: MINIMIZE THE NUMBER OF CALLS TO FUNCTION

** 3 **SET ERRORDEF 0.1852E-01

minuit> *****

** 4 **MIGRAD

FIRST CALL TO USER FUNCTION AT NEW START POINT, WITH IFLAG=4.
START MIGRAD MINIMIZATION. STRATEGY 0. CONVERGENCE WHEN EDM .LT. 0.19E-05

FCN= 688.8897 FROM MIGRAD STATUS=INITIATE 4 CALLS 5 TOTAL
EDM= unknown STRATEGY= 0 NO ERROR MATRIX

EXT	PARAMETER	CURRENT GUESS	STEP	FIRST
NO.	NAME	VALUE	ERROR	SIZE
1	V	38.200	0.10000	0.10000
				-80.749

ERR DEF= 0.185E-01

MIGRAD FAILS TO FIND IMPROVEMENT
MACHINE ACCURACY LIMITS FURTHER IMPROVEMENT.

MIGRAD MINIMIZATION HAS CONVERGED.

FCN= 228.5799 FROM MIGRAD STATUS=CONVERGED 32 CALLS 33 TOTAL
EDM= 0.90E-05 STRATEGY=0 ERROR MATRIX UNCERTAINTY= 20.5%

EXT	PARAMETER	APPROXIMATE	STEP	FIRST
NO.	NAME	VALUE	ERROR	SIZE
1	V	46.737	0.42282E-01	-0.0000
				0.12874E-01

ERR DEF= 0.185E-01

EXTERNAL ERROR MATRIX. NDIM= 50 NPAR= 1 ERR DEF= 0.185E-01
0.179E-02

ERR MATRIX APPROXIMATE

minuit> *****

** 5 **END

CALL TO USER FUNCTION WITH IFLAG = 3

END OF MINUIT COMMAND INPUT. RETURN TO USER PROGRAM.

Control is then returned to SFRESCO which prints out the results and also writes file search.plot, suitable for plotting with xmgrace:

```
sfresco> Var 1=V      value 46.736719, step 0.1000, error 0.0423
sfresco>
Dataset      1 <=>
  Angle  Datum  Abs. error Theory    Chi
  10.880 761.00  10.654   1072.1   852.6287
  13.050 759.00  11.385   1010.5   488.0935
.
.
.
143.220 0.41100E-01 0.13152E-02 0.70738E-02 669.3354
146.940 0.36300E-01 0.15609E-02 0.83952E-02 319.6007

Total ChiSq/N = 228.5799 from 228.580
sfresco>
ChiSq/N = 228.580 from 228.580
xmgr file written: search.plot
```

Thus, the new value of V is 46.74 MeV, giving a χ^2 per point value of 228.58, somewhat better than what we started with. N.B. *SFRESCO does not update the input file p12CKD.fri*, you must do this yourself if you wish to continue searching on another variable from the end point of this search, i.e. you must change the value of V to 46.74 in p12CKD.fri before searching on W, say.

Try searching on various combinations of parameters to get the best fit to the data you can. Be aware that some results may be unphysical – large radii and/or small diffuseness values etc. You should be able to get a final result with a χ^2 per point value of about 10 or slightly less. If you are having trouble with the search persistently heading off towards unphysical values for some variables try setting a constraint on the variable(s) involved, i.e. set limits within which the search must remain. This is done as follows:

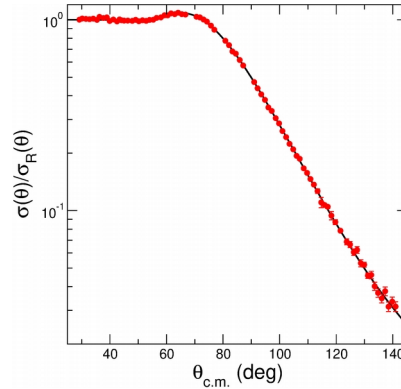
```
&variable kind=1 name='r' step=0.1 kp=1 pline=2 col=2 valmin= 1.0 valmax = 1.5/
```

to keep the real radius parameter between 1.0 and 1.5 fm, for example. If the initial value is outside the range SFRESCO will bring it back inside before commencing the search, so be aware of this.

The final exercise concerns fitting heavy ion data, specifically the 35 MeV $^7\text{Li} + ^{208}\text{Pb}$ elastic scattering we looked at in the previous section. Go to the working directory you created for this example and create a sub-directory called “search”. Copy the files sf.inp and search.in to this directory, along with the file 7Li208Pb.fri. Locate the file nsf35.dat in the materials provided and copy this to the same directory. This file contains the 35 MeV $^7\text{Li} + ^{208}\text{Pb}$ elastic scattering data we are going to fit. The format is: scattering angle, cross section as ratio to the Rutherford cross section, absolute error of the cross section (also expressed as a ratio to the Rutherford cross section).

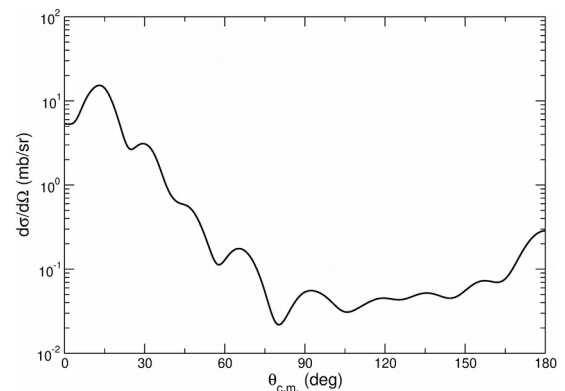
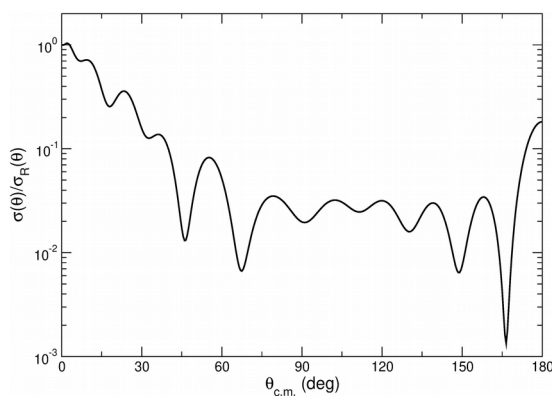
Edit the file search.in and replace the p + ^{12}C data with these data. Remember to change the specification of the data (see Appendix VI) to let SFRESCO know that the cross section is now expressed as ratio to Rutherford. Also change the file names to suit and perform a search on suitable variables. Bearing in mind the expected ambiguities in the heavy-ion optical potential for an energy close to the Coulomb barrier that you will have learned about in the theory lectures, there will be

multiple potentials that give equivalently good fits to the data. Try to search on as few variables as possible to get the best fit – you should find that three is the minimum number to get a truly good fit. You should obtain a cumulative reaction cross section of about 772 mb and obtain a χ^2 per point value of around 1.7. The fit to the data should look something like the following:



3. Coupled Channels Calculations

We will now look at coupled channels calculations for inelastic scattering. In the materials provided, find the file called 7li12c.fri. This is the input discussed in the lectures using TYPE=11 coupling to perform a coupled channels calculation for inelastic scattering of 34 MeV ^7Li from a ^{12}C target leaving the target in its 4.44 MeV 2^+ excited state. Create a new working directory and copy the file there. Run the file and plot the elastic and inelastic scattering angular distributions. These you will find either separately in files fort.201 (the elastic scattering) and fort.202 (the inelastic scattering) or both together in fort.16, the elastic scattering being listed first. The input is so arranged that the elastic scattering is calculated as a ratio to the Rutherford scattering cross section and the inelastic scattering is calculated in mb/sr (both usual for heavy ion reactions of this type). You should see something like this:



You should obtain a total reaction cross section of 1536.48075 mb and a total cross section for the ^{12}C 2^+ state of 8.12523 mb.

The next step is to add coupling to the 14.079 MeV 4^+ state of ^{12}C using TYPE=11 coupling, as per the example in the lectures. Create a new working directory and copy 7li12c.fri to it. Rename it 7li12c_4+.fri and modify appropriately to include k=2 coupling to the 4^+ state. If you plot the new elastic scattering and 2^+ inelastic scattering angular distributions you should notice some differences due to the new couplings to the 4^+ state. If you wish to plot the angular distribution for excitation of the ^{12}C 4^+ state you will find it in file fort.203 or as the third listing in fort.16.

The next exercise is more challenging. Create a sub directory and copy 7li12c_4+.fri to it. Rename it as 7li12c_4+_TYPE13.fri. Following the steps in the lectures, modify the coupling to

TYPE=13. If you do everything correctly you will find that you obtain exactly the same result as with TYPE=11, since for the sake of this exercise all the couplings strengths in the TYPE=13 input are evaluated according to the rotational model. Part of the modification was already demonstrated in the lectures; you need only work out how to include the $2^+ \leftrightarrow 4^+$ and $4^+ \leftrightarrow 4^+$ couplings. If you take your time and follow the example from the lectures it should not prove too difficult. Two quantities you may find useful are the following Clebsch-Gordon coefficients:

$$\langle 2\ 0\ 2\ 0 | 4\ 0 \rangle = 0.717137, \langle 4\ 0\ 2\ 0 | 2\ 0 \rangle = 0.534522 \text{ and } \langle 4\ 0\ 2\ 0 | 4\ 0 \rangle = -0.509647.$$

The main output file also provides useful checks. You should see something like this for the Coulomb part of the potential:

```

1      0=Coulomb      0=CHARGE (WS) 1      A#1      A#2      r0c      ac      0.0000      0.0000      0.0000      0.0000      74.213      h @ 1
                                     Mn(E1)      Mn(E2)      Mn(E3)      .....
1      13=Targ. table 10=derivatives 2      0.0000      -6.3008      0.0000      0.0000      0.0000      0.0000      0.0000      5.253= RADIUS

Coulomb reduced matrix element 1 <- 2 of k = 2 (at 2 from 1 p) == -6.3008
Coulomb reduced matrix element 2 <- 1 of k = 2 (at 2 from 1 p) == -6.3008
Coulomb reduced matrix element 2 <- 2 of k = 2 (at 2 from 1 p) == 7.5309
Coulomb reduced matrix element 2 <- 3 of k = 2 (at 2 from 1 p) == -10.1038
Coulomb reduced matrix element 3 <- 2 of k = 2 (at 2 from 1 p) == -10.1038
Coulomb reduced matrix element 3 <- 3 of k = 2 (at 2 from 1 p) == 9.6336
Step 2 <- 1 of k = 2 (Potl. beta ==-0.15941) 1 @ 2
Beta of targ. ==-0.53710 (using radius of 2.862 fm.).
Step 1 <- 2 of k = 2 (Potl. beta ==-0.15941) 1 @ 2
Beta of targ. ==-0.53710 (using radius of 2.862 fm.).
Step 2 <- 2 of k = 2 (Potl. beta ==-0.15941) 1 @ 2
Beta of targ. ==-0.53710 (using radius of 2.862 fm.).
Step 3 <- 2 of k = 2 (Potl. beta ==-0.15941) 1 @ 2
Beta of targ. ==-0.53711 (using radius of 2.862 fm.).
Step 2 <- 3 of k = 2 (Potl. beta ==-0.15941) 1 @ 2
Beta of targ. ==-0.53711 (using radius of 2.862 fm.).
Step 3 <- 3 of k = 2 (Potl. beta ==-0.15941) 1 @ 2
Beta of targ. ==-0.53711 (using radius of 2.862 fm.).

```

and this for the nuclear part:

```

1      1=Volume      0=Woods-Saxon 3      147.0000      0.6300      0.7300      6.6000      1.3800      0.8500      0.000      74.213
                                     Def1      Def2      Def3      .....
1      13=Targ. table 11=quadrat,k>0 4      0.0000      -1.4000      0.0000      0.0000      0.0000      0.0000      0.000      2.783= RADIUS

Nuclear reduced matrix element 1 <- 2 of k = 2 (at 4 from 3 p) == -1.4000
Nuclear reduced matrix element 2 <- 1 of k = 2 (at 4 from 3 p) == -1.4000
Nuclear reduced matrix element 2 <- 2 of k = 2 (at 4 from 3 p) == 1.6733
Nuclear reduced matrix element 2 <- 3 of k = 2 (at 4 from 3 p) == -2.2450
Nuclear reduced matrix element 3 <- 2 of k = 2 (at 4 from 3 p) == -2.2450
Nuclear reduced matrix element 3 <- 3 of k = 2 (at 4 from 3 p) == 2.1405
Step 2 <- 1 of k = 2 (Potl. beta ==-0.50307) 1 @ 4
Beta of targ. ==-0.92341 (using radius of 1.516 fm.).
Step 1 <- 2 of k = 2 (Potl. beta ==-0.50307) 1 @ 4
Beta of targ. ==-0.92341 (using radius of 1.516 fm.).
Step 2 <- 2 of k = 2 (Potl. beta ==-0.50306) 1 @ 4
Beta of targ. ==-0.92340 (using radius of 1.516 fm.).
Step 3 <- 2 of k = 2 (Potl. beta ==-0.50307) 1 @ 4
Beta of targ. ==-0.92341 (using radius of 1.516 fm.).
Step 2 <- 3 of k = 2 (Potl. beta ==-0.50307) 1 @ 4
Beta of targ. ==-0.92341 (using radius of 1.516 fm.).
Step 3 <- 3 of k = 2 (Potl. beta ==-0.50307) 1 @ 4
Beta of targ. ==-0.92340 (using radius of 1.516 fm.).

```

You should find that the “Beta of targ.” are the same for all transitions for each of the Coulomb and nuclear parts.

We will now progress to a slightly more complicated case, coupling to the $1/2^-$ first excited state of the ${}^7\text{Li}$ projectile and reorientation of its $3/2^-$ ground state, again assuming for the sake of the exercise that all coupling strengths follow the rotational model (a rather good approximation in this case). We will also assume that the $3/2^-$ and $1/2^-$ states form the first two members of an (inverted) $K = 1/2$ band.

Add the $1/2^-$ state as channel number 4, making sure to use COPYt for the target (^{12}C) level, which should be the 0^+ ground state; in these exercises we will not be considering mutual excitation of the projectile and target. We have not thus far mentioned the use of the variable K on card 7. This can be used to define the K quantum number of the band: if it is left blank FRESKO will set a value automatically and this is usually the physical one. Occasionally the default value is incorrect (the main output file will tell you the value used by FRESKO) so it is useful to be able to set it “by hand” if necessary. In this case the default value is $K = 1/2$, as required. While it would be possible to use TYPE=10 coupling for this example we shall use TYPE=12 to give you more practice in using this more flexible form of coupling. We will now give the information necessary to assemble the required input.

Take the excitation energy of the $^7\text{Li } 1/2^-$ state as 0.4776 MeV and the $B(E2; 3/2^- \rightarrow 1/2^-) = 8.3 \text{ e}^2\text{fm}^4$. For the nuclear coupling, take $\delta_2 = 2.0 \text{ fm}$, i.e. $\text{DEF}(2) = 2.0 \text{ fm}$ (Card 11). You may also find the following two Clebsch-Gordon coefficients useful: $\langle 3/2 \ 1/2 \ 2 \ 0 | 1/2 \ 1/2 \rangle = -0.447214$ and $\langle 3/2 \ 1/2 \ 2 \ 0 | 3/2 \ 1/2 \rangle = -0.447214$. Remember that $K = 1/2$ and that a level with spin $1/2$ has no reorientation coupling. If you have included the couplings correctly you should obtain a cumulative reaction cross section of 1527.04342 mb, with integrated cross sections for population of the $^{12}\text{C } 2^+$ and 4^+ and $^7\text{Li } 1/2^-$ states of 8.93818, 0.07416 and 5.75367 mb respectively. You will also probably notice that there is now a non-zero outgoing cross section in the entrance channel (4.08477 mb); this is due to the reorientation coupling.

You should now be reasonably familiar with the basic inputs to a coupled channels calculation of inelastic scattering with FRESKO. In the next exercise we will change the target to ^{208}Pb to gain some experience of systems where Coulomb excitation is important. We shall retain the ^7Li projectile and assemble an input to calculate the inelastic scattering of ^7Li from a ^{208}Pb target at an incident energy of 35 MeV with the ^7Li projectile left in its 0.478 MeV $1/2^-$ state (the ground state reorientation of ^7Li should also be included, as in the previous exercise).

Copy the file 7Li208Pb.fri that you prepared earlier to a new directory and rename it 7Li208Pb_inel.fri. Now add a channel where the ^7Li is in its $1/2^-$ excited state and the ^{208}Pb target in its ground state. Add the couplings in exactly the same way as you did in the previous exercise (N.B. remember to change IBLOCK on Card 4). You may retain the optical potential parameters from the original optical model calculation.

Run the input file and look critically at the output, in particular the inelastic scattering. It may look slightly odd. The explanation is quite simple. If you look at the very end of the output file you should see something like this:

Elastic $h^*k = 0.199$ so OK compared with 0.200

Real(S-el) > 0.01 first at J =	0.5
Real(S-el) > 0.10 first at J =	1.5
Real(S-el) > 0.50 first at J =	13.5
Real(S-el) > 0.90 first at J =	20.5
Real(S-el) > 0.99 first at J =	30.5
R-turn = 24.21 fm at J =	60.5

Forward-angle excitation cut off below 32.584 deg by max JT = 60.5
and below 24.016 deg by max R.

This means that with only 60 partial waves the Coulomb excitation is cut off for angles smaller than 32.584° , in other words for scattering angles $\theta_{\text{c.m.}} < 32.584^\circ$ the calculated inelastic scattering is not correct. To rectify this, it is necessary to increase the number of partial waves included in the calculation, i.e. to increase parameter JTMAX on Card 2. You will also need to increase RMATCH

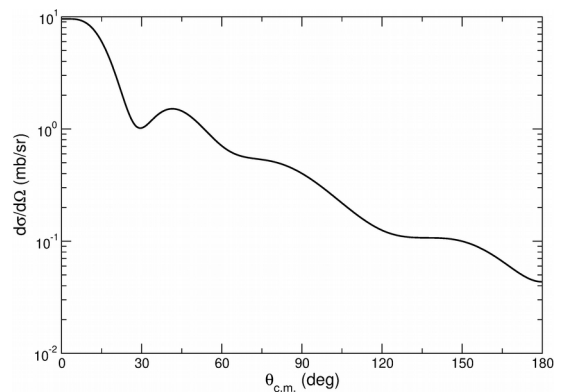
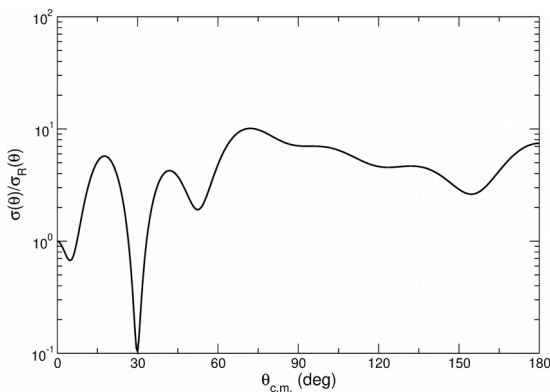
on Card 1 accordingly. Gradually increase the number of partial waves included in your calculation until the output tells you that the cut-off is for angles smaller than about 7° and compare the inelastic scattering with your original result. This will underline the importance of including enough partial waves when calculating the inelastic scattering for systems with strong Coulomb excitation.

As a final example of an inelastic scattering calculation we consider a system with strong coupling, $^{12}\text{C} + ^{152}\text{Sm}$ at an incident energy of 63.2 MeV. Copy your 7Li208Pb_inel.fri input to a new directory and rename it 12C152Sm_inel.fri. Modify it to include coupling to the 0.1218 MeV 2^+ state of ^{152}Sm . Take the mass of ^{152}Sm as 151.9187 and the Mn(E2) and DEF(2) values as 189.4 efm² and 1.67 fm respectively. Include the coupling using TYPE=11 (we assume that the 2^+ state is the first excited level of a K=0 rotational band). Use an optical potential of standard Woods-Saxon form with parameters: $V = 13.0$ MeV, $r_V = 1.41$ fm, $a_V = 0.41$ fm, $W = 4.0$ MeV, $r_W = 1.43$ fm, $a_W = 0.41$ fm and $r_C = 1.25$ fm with the “heavy ion” radius convention, i.e. $R_x = r_x (12^{1/3} + 152^{1/3})$ fm.

To begin with, take JTMAX=300, choosing suitable values of HCM and RMATCH to enable the calculation to run over the full range of partial waves. You should obtain a cumulative reaction cross section of 1994.08157 mb and a cumulative outgoing cross section for the $^{152}\text{Sm } 2^+$ of 1021.72278 mb. You should find that the output tells you that the inelastic excitation is cut off for angles forward of about 10° . Create a new sub-directory and copy 12C152Sm_inel.fri to it and rename as 12C152Sm_inel_3000.fri. Now increase JTMAX to 3000 (three thousand) and, using the information given in the output file, adjust RMATCH accordingly to enable the calculation to run over the full range of partial waves. You should now obtain a cumulative reaction cross section of 2076.41248 mb and a cumulative outgoing cross section for the $^{152}\text{Sm } 2^+$ of 1104.09299 mb. If you plot the elastic and inelastic scattering angular distributions from the two calculations you will find that the two results for the elastic scattering are indistinguishable on a plot whereas the inelastic scattering results differ significantly for angles $\theta_{\text{c.m.}} < 20^\circ$ (they are otherwise indistinguishable apart from a few minor wiggles at large angles). The output file will now inform you that the inelastic excitation is cut off for angles forward of about 1° . This example shows the rather extreme measures required if you wish to calculate an accurate inelastic scattering cross section at small angles for systems where there is strong Coulomb excitation. It also demonstrates that exactly how many partial waves are required can depend very much on which quantities one is interested in; in this case, a reliable calculation of the elastic scattering would only require 300 partial waves (perhaps even less) but if the inelastic cross section at small angles is required many more partial waves will be needed.

4. Distorted Wave Born Approximation Calculations

We now move on to calculations of transfer reactions using the distorted wave Born approximation (DWBA). Find the file 12cdp.fri in the materials provided: this is the input file used as an example in the lectures. Copy to a new working directory and run it with FRESKO. Plot the elastic scattering and transfer angular distributions; they should look like this:



You should obtain a cumulative reaction cross section of 1026.27257 mb and a cumulative outgoing cross section in partition 2 (here the integrated transfer cross section) of 8.65904 mb.

Taking these results as our benchmark, we will now investigate the influence of the choice of RINTP on the calculated transfer cross section (it will not influence the elastic scattering in DWBA). It will be simplest if you create suitably named sub-directories for each value of RINTP. Increase RINTP from 0.2 to 0.3 fm and compare the result for the transfer cross section; it should not be significantly different. Further increase RINTP to 0.4 and 0.5 fm. You will find that the calculated transfer cross section becomes progressively smaller as RINTP is increased. Check that the result with RINTP = 0.1 fm is essentially identical to our benchmark result with RINTP = 0.2 fm. We thus see that a “converged” transfer cross section is obtained with any value between 0.1 and 0.3 fm. Use of larger values will lead to increasingly erroneous results. Note that the error induced by the use of too large a value of RINTP is not obvious simply by looking at the calculated angular distribution; it seems perfectly reasonable.

If you perform a similar exercise for HNL (with RINTP fixed at our benchmark value of 0.2 fm) you will find that the calculated transfer cross section is much less sensitive to this variable, any value between 0.1 and 0.5 fm giving essentially the same result. The variation as a function of RNL is even slower, there being no real change in the calculated transfer cross section for values of RNL between 4 fm (about the smallest value FRESKO will allow in this case) and at least 20 fm. Since the calculation time can vary greatly depending on the choice of these variables (in this simple case calculation time is not an issue, of course) it is in your interest to optimise their values. However, you should always try to establish that your result is converged by checking that reasonable changes do not affect the calculated cross section(s).

Finally, extract the squared ANC, C^2 , for the $^{13}\text{C}(\text{g.s.}) | n + ^{12}\text{C}(\text{g.s.})$ overlap according to the instructions given in the lectures. You should find a value of $C^2 = 2.633 \text{ fm}^{-1}$.

The next exercise is to add transfers to two further states in ^{13}C , the 3.089 MeV $1/2^+$ and the 3.685 MeV $3/2^-$. Create a new directory and copy 12cdp.fri to it. Rename it 12cdp_3.fri. When adding stripping to the two excited states of ^{13}C you may assume the same $n + ^{12}\text{C}$ binding potential parameters as for the ground state and take the $p + ^{13}\text{C}$ optical potentials for the two new channels to be of the same form as for the ground state but with parameters calculated according to:

$$V_0 = 63.1 - 0.3E_{\text{c.m.}} \text{ MeV}$$

$$r_0 = 1.15 - 0.001E_{\text{c.m.}} \text{ fm}$$

$$a_0 = 0.57 \text{ fm}$$

$$W_D = 10.37 - 0.06E_{\text{c.m.}} \text{ MeV}$$

$$r_D = 1.15 - 0.001E_{\text{c.m.}} \text{ fm}$$

$$a_D = 0.5 \text{ fm}$$

$$V_{so} = 5.5 \text{ MeV}$$

$$r_{so} = 1.15 - 0.001E_{\text{c.m.}} \text{ fm}$$

$$a_{so} = 0.57 \text{ fm}$$

$$r_C = 1.15 - 0.001E_{\text{c.m.}} \text{ fm}$$

where $E_{\text{c.m.}}$ is the energy of the outgoing proton *in the centre of mass frame*. You may take the following values for the spectroscopic factors of the two new overlaps: $C^2S(^{13}\text{C } 3.089) = 0.65$ and $C^2S(^{13}\text{C } 3.685) = 0.14$. You should now have all the information you need to produce the input file – do not forget to use COPY correctly on cards 7 and think carefully about the shell model levels the transferred neutron must populate to form the ^{13}C excited states. The cumulative outgoing cross sections should be 6.83507 mb and 2.06800 mb for stripping to the $1/2^+$ and $3/2^-$ states respectively.

Extract the squared ANC values for the two new states; you should obtain values of $C^2 = 2.157 \text{ fm}^{-1}$ and 0.053 fm^{-1} for the $1/2^+$ and $3/2^-$ states respectively.

5. Putting it All Together

We have thus far only considered so-called stripping reactions. In the lectures we mentioned some of the particularities of pickup reactions but we did not give any specific examples. In this exercise you will put into practice most of what you have learned about DWBA reactions by converting the $^{12}\text{C}(\text{d},\text{p})^{13}\text{C}$ stripping input into one for $^{12}\text{C}(\text{p},\text{d})^{11}\text{C}$ pickup.

Create a new directory and copy the file 12cdp.fri to it. Re-name the file 12cpd.fri. As a first step, convert cards 6 and 7 for the entrance and exit partitions to $\text{p} + ^{12}\text{C}$ and $\text{d} + ^{11}\text{C}$. You will need the mass of ^{11}C , which is 11.0114 and the $^{12}\text{C}(\text{p},\text{d})^{11}\text{C}$ Q value, -16.4961 MeV . We will consider pickup leading to just the ground state of ^{11}C , which has spin-parity of $3/2^-$. Now change the optical potentials for entrance and exit channels as appropriate. Take the incident energy of the protons as 51.93 MeV (this value was chosen because it allows us to compare with actual data for this reaction). Initially we shall use global parameter sets adapted for p-shell targets. To save time the potential parameters are as follows:

protons

$$V = 46.68 \text{ MeV}$$

$$r_0 = 1.102 \text{ fm}$$

$$a_0 = 0.57 \text{ fm}$$

$$W = 7.50 \text{ MeV}$$

$$r_W = 1.102 \text{ MeV}$$

$$a_W = 0.50 \text{ fm}$$

$$W_D = 6.73 \text{ MeV}$$

$$r_D = 1.102 \text{ fm}$$

$$a_D = 0.50 \text{ fm}$$

$$V_{SO} = 5.50 \text{ MeV}$$

$$r_{SO} = 1.102 \text{ fm}$$

$$a_{SO} = 0.57 \text{ fm}$$

$$r_C = 1.102 \text{ fm}$$

deuterons

$$V = 89.53 \text{ MeV}$$

$$r_0 = 1.053 \text{ fm}$$

$$a_0 = 0.776 \text{ fm}$$

$$W = 10.7 \text{ MeV}$$

$$r_W = 1.874 \text{ MeV}$$

$$a_W = 0.744 \text{ fm}$$

$$W_D = 0.108 \text{ MeV}$$

$$r_D = 1.874 \text{ fm}$$

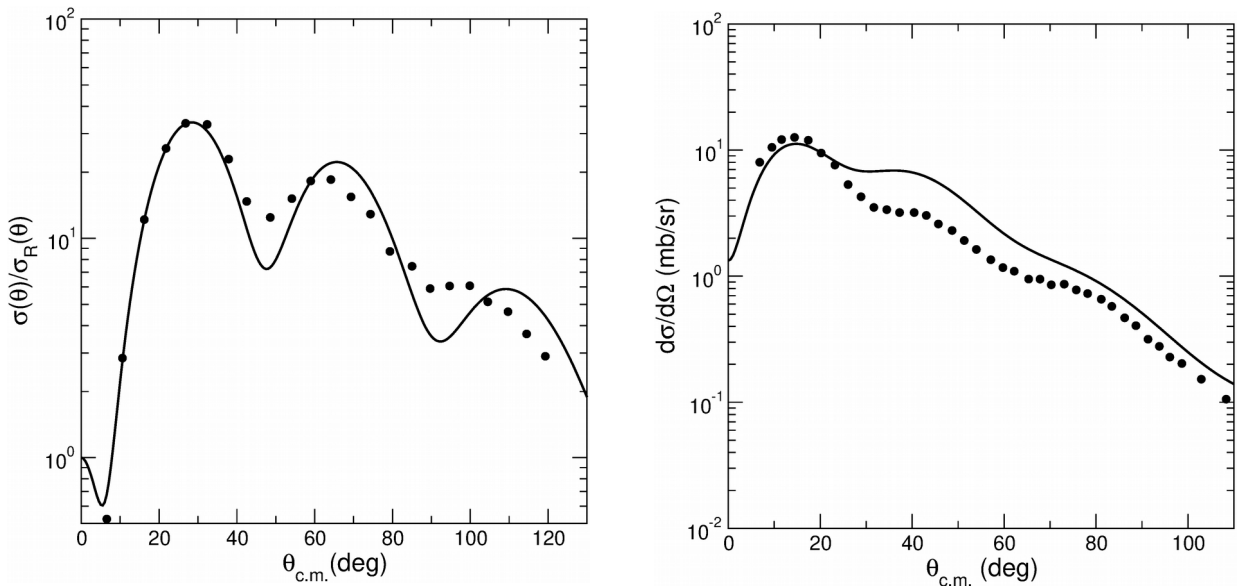
$$a_D = 0.744 \text{ fm}$$

$$r_C = 1.30 \text{ fm}$$

The $\langle d | n + p \rangle$ overlap is the same as for the (d,p) case, so you will not need to change it. However, for the target-like overlap we now need that for $\langle {}^{12}\text{C} | n + {}^{11}\text{C} \rangle$ rather than $\langle {}^{13}\text{C} | n + {}^{12}\text{C} \rangle$. Think about what this means in terms of the “core” and “composite” nuclei and what will need to be changed in the definition of the overlap (including the $n + \text{core}$ binding potential; you may use the same r_0 and a_0 values as for the stripping case). As the name implies, in a pickup reaction we pick up a particle (here a neutron) from the target nucleus and transfer it to the projectile. Thus, in this specific case we want to pick up a neutron from the ground state of ${}^{12}\text{C}$ to leave a ${}^{11}\text{C}$ in its $3/2^-$ ground state. Consider the most likely shell model level the picked up neutron would have been in and from this deduce the N , L and J values required. Also consider the binding energy of the picked up neutron; again, think carefully about the definition of core and composite nuclei in this context. Hint: for pickup the target is the composite nucleus and the target-like residue the core nucleus. You may find the following values of S_n useful: $S_n({}^{13}\text{C}) = 4946.31 \text{ keV}$, $S_n({}^{12}\text{C}) = 18720.71 \text{ keV}$, $S_n({}^{11}\text{C}) = 13120.59 \text{ keV}$.

The definition of the couplings should be reasonably straightforward, although remember that this is pickup and what this implies about the use of the POST or PRIOR form of DWBA. Finally, you will need a spectroscopic amplitude for the $\langle {}^{12}\text{C} | n + {}^{11}\text{C} \rangle$ overlap. We will start with the value calculated by Cohen and Kurath, Nucl. Phys. A **101**, 1 (1967), $S = 5.6989$. You may find the following isospin Clebsch-Gordan coefficient useful: $\langle 1/2, -1/2; 1/2, 1/2 | 0, 0 \rangle = 1/\sqrt{2}$ (recall FRESKO’s definition of the spectroscopic factor); $T({}^{12}\text{C} \ 0.0 \text{ MeV}) = 0$ and $T({}^{11}\text{C} \ 0.0 \text{ MeV}) = 1/2$.

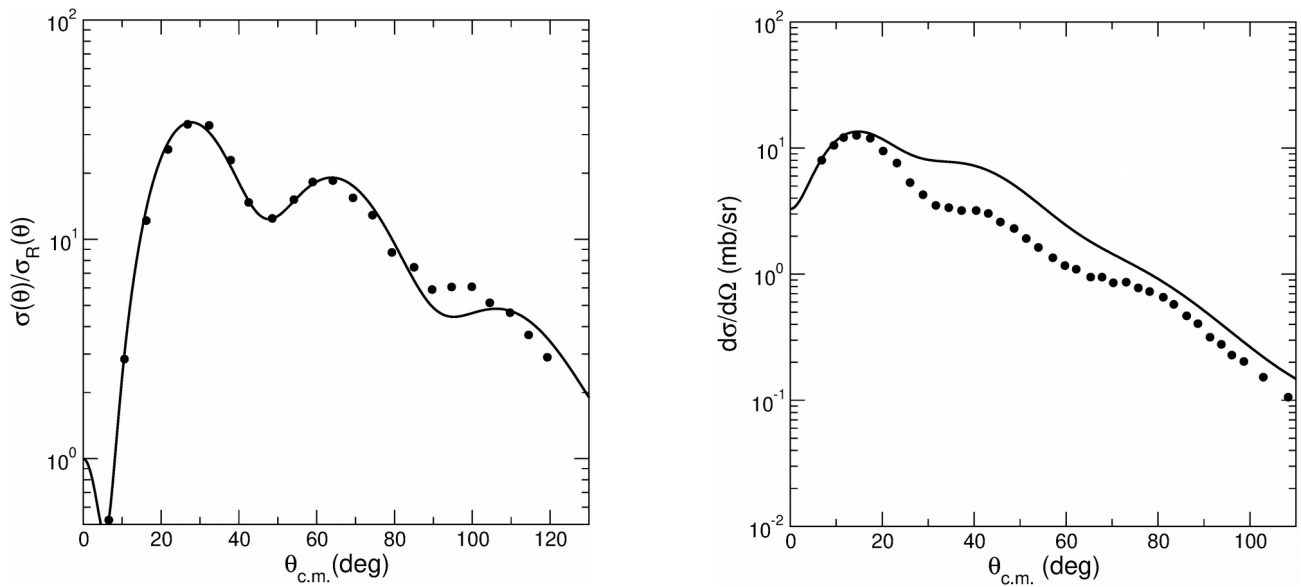
When you have put everything together, run FRESKO using your input. You may need to adjust some of the variables on card 1 such as HNL, RNL etc. If all goes well you should obtain a cumulative reaction cross section of 369.53437mb and an outgoing cross section for the pickup of 23.03006 mb. You will find among the materials data for the elastic scattering (expressed as a ratio to the appropriate Rutherford scattering cross section) and the pickup to the ${}^{11}\text{C} \ 3/2^-$ ground state and if you plot your calculations together with these you should find something like the following:



The description of the elastic scattering data is reasonable given that we are using a set of global optical potential parameters, albeit specially adapted for p-shell targets. The description of the pickup could be better; not only is the shape not particularly well described but the magnitude of the

forward angle peak is slightly under-predicted. This latter at least could be remedied by a slight increase in the spectroscopic factor. However, we will investigate the effect on the description of the pickup cross section of improving the description of the entrance channel elastic scattering.

You should by now be able to produce an elastic scattering input (cut down from your DWBA input file) and arrange the sf.inp and search.in files with the data so that you can search on the parameters of the $p + {}^{12}\text{C}$ optical potential. Recall that the data are given as a ratio to the Rutherford scattering cross section, so this will need to be borne in mind when arranging the file search.in. You should be able to improve the description of the data to the extent that the nominal χ^2 per point is reduced to a value of about 37 (the absolute value is not physically meaningful here since the data were given a nominal uncertainty of $\pm 2\%$ at all angles). If you then perform a DWBA calculation using your new parameters for the entrance channel optical potential you should obtain a result similar to this:



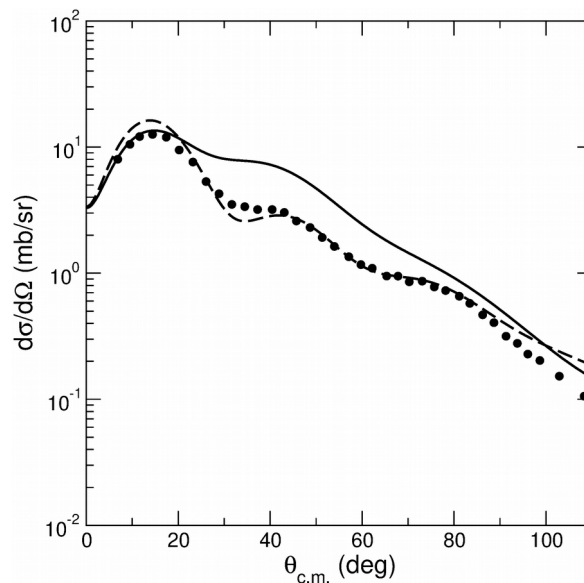
The improved description of the elastic scattering has had little impact on the shape of the pickup angular distribution but it has increased the magnitude to match the forward angle peak. For the exit channel $d + {}^{11}\text{C}$ optical potential using the global parameters is probably the best we can do, since there are no $d + {}^{11}\text{C}$ elastic scattering data available.

However, there is one important property of the deuteron that we have overlooked in our analysis. It is relatively weakly bound (the threshold against breakup into a neutron and a proton is 2.2246 MeV) and so can quite easily break up into a proton and neutron. This will have an influence on the (p,d) cross section (and also, incidentally, on the (d,p) one too) which is not properly taken into account in the DWBA. There are various more or less sophisticated ways of handling the deuteron breakup in this context, but there is a simplified theory, known as the adiabatic model, that works very well. Basically, in this model the reaction theory is essentially identical to standard DWBA with the exception that the deuteron distorted waves are redefined and are no longer those associated with the deuteron elastic scattering. This is achieved by replacing the usual deuteron optical potential by the so-called adiabatic potential in an otherwise standard DWBA calculation.

The adiabatic potential may be calculated in an approximate way from the neutron and proton optical potentials for half the incident deuteron energy using formulae given in J. D. Harvey and R. C. Johnson, Phys. Rev. C **3**, 636 (1971). If we carry out this exercise for the case under discussion here using the same set of global nucleon optical potential parameters as we used to calculate the entrance channel potential we obtain the following set of parameters for the $d + {}^{11}\text{C}$ adiabatic potential:

$V = 110.9 \text{ MeV}$
 $r_0 = 1.133 \text{ fm}$
 $a_0 = 0.61 \text{ fm}$
 $W_D = 14.71 \text{ MeV}$
 $r_D = 1.133 \text{ fm}$
 $a_D = 0.54 \text{ fm}$
 $V_{SO} = 5.50 \text{ MeV}$
 $r_{SO} = 1.133 \text{ fm}$
 $a_{SO} = 0.57 \text{ fm}$
 $r_C = 1.133 \text{ fm}$

If you replace the exit channel potential with these parameters while retaining the fitted entrance channel potential you should obtain something like the dashed curve in the figure below (the solid curve is the previous DWBA result for comparison):



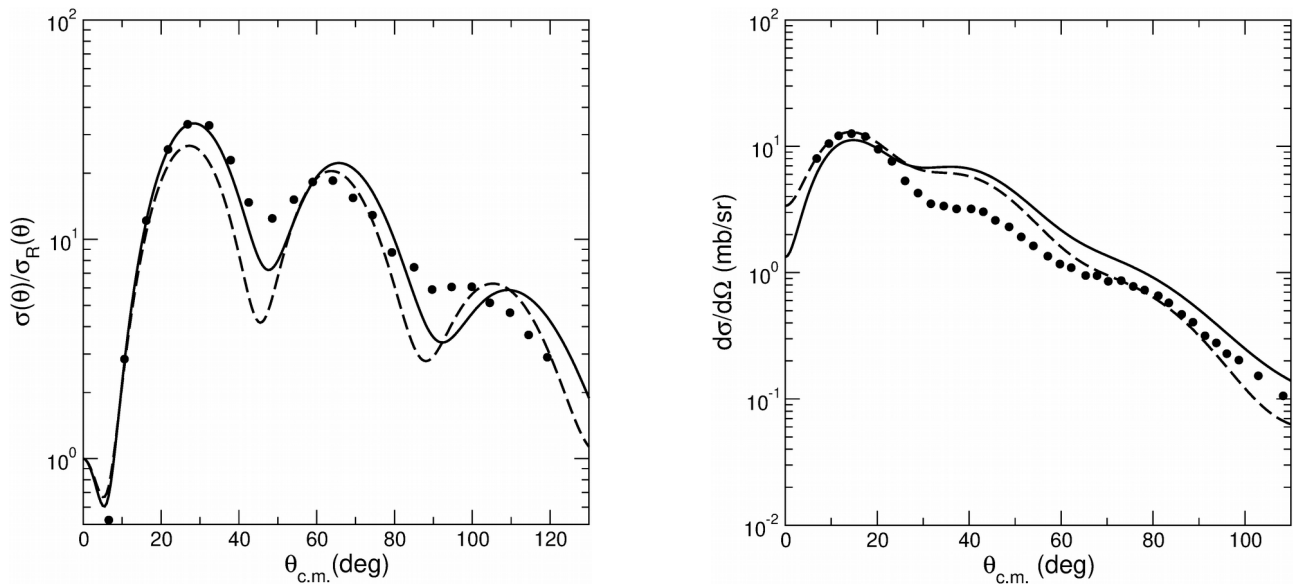
The description of the shape is much closer to the data, although the forward angle peak is now a little over-predicted (by approximately 20%). This illustrates the potential importance of taking account of deuteron breakup effects in (p,d) and (d,p) calculations.

6. Coupled Reaction Channels Calculations

You should now be reasonably confident in how to assemble a DWBA input for typical light-ion induced reactions. In this exercise we will briefly look at how to convert a DWBA input into a coupled reaction channels (CRC) one. Copy the file 12cdp_3.fri to a new directory and rename it 12cdp_3_crc.fri. Following the example in the lectures, convert it from DWBA to CRC. If it does not run at first, look at the output file and consider carefully what it tells you. In particular, check the values of RNL and CENTRE on card 1. You should find that in this case the transition from DWBA to CRC makes little difference to the calculated angular distributions, since here not only are the transfer couplings rather weak but also we have not included any multi-step reaction paths. However, it does not necessarily follow from this that DWBA is an adequate theory in this

case, since in reality to describe the available data for this reaction it is necessary to introduce more complex reaction paths that can only be handled adequately with CRC.

Now perform a similar exercise for 12cpd.fri. Again, you may need to change some of the parameters on Card 1 such as RNL and CENTRE. You will probably find that even with this the input refuses to run to the end with a reasonable maximum number of iterations ITER on Card 4. This is because here the transfer couplings are strong and the iterative series diverges (or converges very slowly). This can be remedied by setting the parameter PADE on Card 4 equal to 1. This invokes Padé acceleration, a mathematical trick which obviates this problem. If you start from the input file using the global optical potential parameters for both entrance and exit channels you should find something like the following when comparing the CRC angular distributions (dashed curves) with the DWBA ones (solid curves):



In this case, although the coupling is strong and has had a considerable effect on the elastic scattering the influence on the pickup cross section is smaller (but not insignificant in the context of extracting a spectroscopic factor since it increases the forward angle peak cross section by about 15%). In principle one should search on the entrance channel optical potential parameters to recover the description of the elastic scattering but since this is somewhat more time consuming in the context of a CRC calculation than for a simple optical model calculation we will leave this as an “exercise for the reader” for those who are curious as to what influence this may have on the pickup cross section. In fact, while it is possible to obtain a good description of the elastic scattering in this way the accompanying description of the pickup is not improved.

7. Heavy ion Induced Transfer Reactions and Cores with Non-zero Spin

In this section we will consider the slight complications introduced into transfer reaction calculations when either (or both) of the cores has an intrinsic spin different from zero. The (d,p) and (p,d) reactions present very specific cases of this type via the $\langle d | n + p \rangle$ overlap but here we illustrate a more general method of dealing with such situations via the $^{208}\text{Pb}(^7\text{Li}, ^6\text{Li})^{209}\text{Pb}$ single-neutron stripping reaction. This will also give you some experience of reactions induced by heavy ions.

We shall keep to an incident ^7Li energy of 35 MeV, as in our optical model example. You may either modify the original optical model input file or start with one of the transfer input files, as you find more convenient. The details for the entrance channel are as for the previous $^7\text{Li} + ^{208}\text{Pb}$ optical model example. For the exit partition you will need the masses of ^6Li (6.0151) and ^{209}Pb

(208.9811) as well as the Q value for the $^{208}\text{Pb}(^7\text{Li}, ^6\text{Li})^{209}\text{Pb}$ reaction (-3.3137 MeV). We will include stripping of a neutron from the ^7Li $3/2^-$ ground state to the ^6Li 1^+ ground state leaving the residual ^{209}Pb nucleus in the following five states: 0.0 MeV $9/2^+$, 1.57 MeV $5/2^+$, 2.03 MeV $1/2^+$, 2.49 $7/2^+$ and 2.54 MeV $3/2^+$. The exit partition will thus contain five channels. Do not forget to use COPY as appropriate on Cards 7 when defining these channels.

For the entrance channel optical potential, take the same parameters as in the initial optical model exercise for the $^7\text{Li} + ^{208}\text{Pb}$ system, i.e. those of the global set of Cook. In the same article Cook also gives a set of global parameters for ^6Li and we will use these for the exit channel optical potential (the same for all channels since the parameter set of Cook is independent of incident energy). These are given by:

$$\begin{aligned} V_R &= 109.5 \text{ MeV}, \quad r_0 = 1.326 \text{ fm}, \quad a_0 = 0.811 \text{ fm}, \\ W_S &= 58.16 - 0.328A + 0.00075A^2 \text{ MeV}, \quad r_I = 1.534 \text{ fm}, \quad a_I = 0.884 \text{ fm}, \\ r_C &= 1.30 \text{ fm} \end{aligned}$$

with the same radius convention as for the ^7Li set.

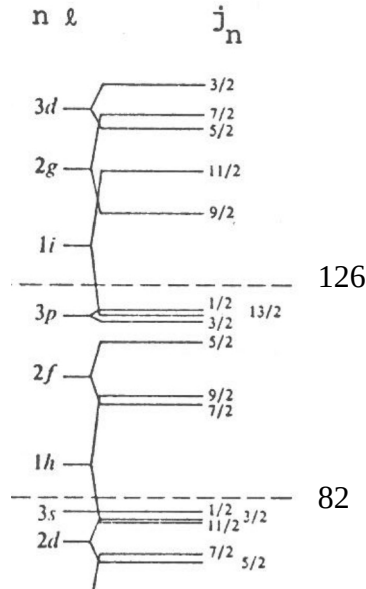
For the potentials binding the transferred neutron to the cores, use $r_0 = 1.25$ fm, $a_0 = 0.65$ fm plus a spin-orbit term with $V_{SO} = 6.0$ MeV, $r_{SO} = 1.25$ fm, $a_{SO} = 0.65$ fm for the $\langle ^7\text{Li} | n + ^6\text{Li} \rangle$ overlap and $r_0 = 1.23$ fm, $a_0 = 0.65$ fm plus a spin-orbit term with $V_{SO} = 6.0$ MeV, $r_{SO} = 1.23$ fm, $a_{SO} = 0.65$ fm for the $\langle ^{209}\text{Pb} | n + ^{208}\text{Pb} \rangle$ overlaps (you should define your overlaps so that FRESKO sets the central well depths to give the correct binding energies automatically; use $V = 50$ MeV as your starting value in the definition of the binding potentials). To define the overlaps, you may take $S_n(^7\text{Li}) = 7.2511$ MeV and $S_n(^{209}\text{Pb}) = 3.9374$ MeV.

We now come to the complication, defining the N, L and J quantum numbers for the $\langle ^7\text{Li} | n + ^6\text{Li} \rangle$ overlap. In our angular momentum algebra we now have to take account of the $J_c = 1$ of the ^6Li core. We need to couple a neutron to the ^6Li core in its 1^+ ground state to produce a ^7Li in its $3/2^-$ ground state. Looking at a standard shell model scheme, one would naturally assume that the neutron would go into the $1p_{3/2}$ orbital, i.e. it would have $N = 1$, $L = 1$ and $J = 3/2$. This will certainly give us a ^7Li in a $3/2^-$ state: first of all, the parity is correct, $(-1)^1 \times (+1) = -1$ where the first term derives from the neutron and the second term from the ^6Li core. Now the angular momentum coupling: we must couple the J of the neutron to the J of the core (^6Li) to give us the J of the composite (^7Li), i.e. the coupling $3/2 + 1$ which can give us $J(^7\text{Li}) = 5/2, 3/2$ or $1/2$. So far, so good. However, there is another possibility. The transferred neutron could equally well be in a $1p_{1/2}$ orbital, since the parity is still good ($L = 1$ here too) and $1/2 + 1$ can give us $J(^7\text{Li}) = 3/2$ or $1/2$.

We thus see that we have two components to the $\langle ^7\text{Li} | n + ^6\text{Li} \rangle$ overlap, one where the neutron has $N = 1$, $L = 1$ and $J = 3/2$ and one where it has $N = 1$, $L = 1$ and $J = 1/2$. We therefore need to define this overlap twice, once for each component, each with its own KP value and specific spectroscopic amplitude A (on Card 17). This procedure assumes no mixing between the two components. The only real difficulty is in defining the values of A. It is a practical impossibility to do this empirically, so we have to fall back on the results of a suitable structure calculation. In this case, it has been found that the values of A for the two components calculated by Cohen and Kurath give physically reasonable results, so we will use these values: $A(1p_{1/2}) = -0.5379$ and $A(1p_{3/2}) = 0.6567$. Note that the relative negative sign has been established by comparison of calculated and measured analysing powers, although in this DWBA calculation it does not affect the results.

The $\langle ^{209}\text{Pb} | n + ^{208}\text{Pb} \rangle$ overlaps are straightforward, since the ^{208}Pb core has spin-parity 0^+ . Defining the N, L and J values for each of the five levels in ^{209}Pb should be relatively easy if the following fragment of the standard shell model scheme for neutrons above the $N = 126$ shell closure is consulted. Population of the various unoccupied levels above $N = 126$ will naturally yield the

various levels of ^{209}Pb that we are concerned with. For example, to form the $9/2^+$ ground state of ^{209}Pb we naturally place the transferred neutron in the $2g_{9/2}$ level.



The final inputs required are the values of the spectroscopic factors for these overlaps. These are 0.92, 0.89, 0.85, 0.95 and 0.99 for the 0.0 MeV $9/2^+$, 1.57 MeV $5/2^+$, 2.03 MeV $1/2^+$, 2.49 $7/2^+$ and 2.54 MeV $3/2^+$ levels respectively. You should now have all the information necessary to assemble a DWBA input file. Remember to check the values of the parameters on Card 1, particularly RNL and CENTRE. If all is correct, you should obtain values for the cumulative outgoing cross sections for populating each of the five ^{209}Pb states of 8.01542 mb, 3.96650 mb, 0.57566 mb, 1.75055 mb and 1.04922 mb respectively.

8. Cluster Transfer Reactions

In this final section we shall look briefly at cluster transfer reactions, specifically α cluster transfers since these will probably be of most interest to you. We will keep these final exercises simple. Find the input file 58ni7lit.fri in the materials provided. This is the example used to illustrate α cluster transfer in the lectures. Run the file and obtain cross section angular distributions for transfer to the 0^+ and 2^+ levels of ^{62}Zn . Now create a sub-directory and copy the input file to it. Edit the file and reduce the values of NN for these two states on Cards 13 by 1, i.e. set NN = 6 and 5 respectively. Run the file again and compare the new cross sections for these levels with the original ones. You will find that the main effect of reducing NN is to reduce the magnitude of the cross sections (by a factor of approximately 1.8 in both cases), with some slight changes in the shapes of the angular distributions (these latter are, however, so slight that it would be practically impossible to measure them). This illustrates the importance of defining NN correctly, or at least of giving the values used in your work when you come to write it up!

One can also, at least in principle, give the ANC for a cluster transfer. However, for transfer of charged particles, in particular α particles, the numbers obtained can be inconveniently large. Thus, the concept of the Coulomb renormalised ANC was introduced, defined as [A. M. Mukhamedzhanov, Phys. Rev. C **86**, 044615 (2012)]:

$$\tilde{C} = \frac{\ell!}{\Gamma(\ell + 1 + \eta)} C,$$

where C is the usual ANC that we met during the lectures (and which is calculated by FRESKO), η is the Sommerfeld parameter for the relative motion of the cluster-core system (also calculated by FRESKO and given in file fort.46), ℓ is the angular momentum of the transferred particle relative to the core and Γ is the gamma function.

Let us now go through the steps required to obtain the Coulomb renormalised ANC for the $\langle {}^{62}\text{Zn}(0^+) | \alpha + {}^{58}\text{Ni} \rangle$ overlap. Firstly, we obtain the usual ANC, as calculated by FRESKO. If you open file fort.46 you should see something like this:

```
24.900 -3.70570 -0.235755E-04 0.636195E-05 -0.463757 0 0.592901E-05
# For Channel # 2: l = 0, j = 0.0, c# 1; k,eta = 0.77654 9.29281
0.540 0.212738E+08 0.348175 0.163663E-07 -4.82844 0 3.37267
0.600 0.212744E+08 0.262409 0.123345E-07 -4.60504 0 1.20927
.
.
.
```

Scroll down until you see these lines:

```
24.840 0.264093E+14 0.268687E-10 0.101740E-23 -1.08332 0 0.760660E-23
24.900 0.264093E+14 0.251783E-10 0.953387E-24 -1.08269 0 0.709936E-23
```

As per the example in the lectures, the ANC is $0.264093\text{E}+14 \text{ fm}^{-1/2}$. This is the “single particle ANC”, so the actual ANC we require is: $C = 0.264093\text{E}+14 \times 0.3241 = 0.855925413\text{E}+13 \text{ fm}^{-1/2}$ where 0.3241 is the relevant spectroscopic *amplitude* A from Card 17. We now need the argument of the gamma function, $\ell + 1 + \eta$. Here, $\ell = 0$ and $\eta = 9.29281$ (taken from file fort.46) so that we need $\Gamma(10.29281) = 704,769.4302405$. You may obtain gamma function values from tables of mathematical functions or from a convenient on-line calculator:

<https://keisan.casio.com/exec/system/1180573444>

We therefore have that the Coulomb normalised ANC is $0.855925413\text{E}+13/704,769.4302405 = 1.21 \times 10^7 \text{ fm}^{-1/2}$, so that the squared value (as normally given) is $1.47 \times 10^{14} \text{ fm}^{-1}$. Carry out a similar exercise for the 2^+ level. You should obtain a value for the squared Coulomb normalised ANC of $1.88 \times 10^9 \text{ fm}^{-1}$.

We have now come to the end of the exercises. Hopefully they will have helped you assimilate and put into practice what you have learned in the lectures. FRESKO is a very flexible code that can do many more things than we have considered in this brief introduction, but you should now have a basis on which to build as and when you need to do more complex calculations.